

Module Title: Stochastic Calculus for Finance

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School of Mathematics

Semester Two 202324

Calculator instructions:

- You are allowed to use a non-programmable calculator in this examination.

Dictionary instructions:

- You are not allowed to use your own dictionary in this exam. A basic English dictionary is available to use. Raise your hand and ask an invigilator if you need it.

Exam information:

- There are 4 pages to this examination.
- There will be **2 hours 30 minutes** to complete this examination.
- This examination is worth 80% of the module mark.
- There are 3 questions in this exam paper. You must answer all 3 questions.
- Questions 1 and 3 count for 30 marks while Question 2 counts for 40 marks.
- The numbers in brackets indicate the marks available for each sub-question.
- You must show all your calculations.
- You must write your answers in the answer booklet provided. If you require an additional answer booklet, raise your hand so an invigilator can provide one.
- You must clearly state your name and Student ID Number in the relevant boxes on your answer booklet. Other boxes may be left blank.

1. (a) [8 marks] Let Ω be a set. Give the definition of a σ -algebra $\mathcal{F}$ on Ω .
- (b) [6 marks] Let $\Omega = \{1, 2, 3, 4, 5, 6\}$. Consider the subset $\{2, 4\} \subset \Omega$. Denote by $\mathcal{F}$ the σ -algebra generated by $\{2, 4\}$, that is $\mathcal{F} = \sigma(\{2, 4\})$. List all the elements of $\mathcal{F}$.
- (c) [8 marks] Consider the measurable space $(\Omega, \mathcal{F})$ introduced in part (b), and let $\mathcal{B}(\mathbb{R})$ denote the Borel σ -algebra on $\mathbb{R}$. Consider two functions $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ given by

$$X(\omega) = \begin{cases} 1, & \text{if } \omega \in \{2, 4\} \\ 2, & \text{otherwise,} \end{cases}$$

and

$$Y(\omega) = \begin{cases} 5, & \text{if } \omega \in \{2, 4, 6\} \\ 10, & \text{otherwise.} \end{cases}$$

- i. Is X $\mathcal{F}/\mathcal{B}(\mathbb{R})$ -measurable?
ii. Is Y $\mathcal{F}/\mathcal{B}(\mathbb{R})$ -measurable?

Provide arguments justifying your answers.

- (d) [8 marks] Let $(\Omega, \mathcal{F})$ be a measurable space.

- i. Give the definition of a probability measure $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$.
ii. By using the defining properties of $\mathbb{P}$, show that for any $A, B \in \mathcal{F}$, we have

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $T > 0$, and consider a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ consisting of σ -algebras $\mathcal{F}_t \subset \mathcal{F}$.

- (a) [5 marks] Give the definition of a one dimensional Wiener process $(W(t))_{t \in [0, T]}$ on $(\Omega, \mathcal{F}, \mathbb{P})$. When do we say that a Wiener process $(W(t))_{t \in [0, T]}$ on $(\Omega, \mathcal{F}, \mathbb{P})$ is an $\mathbb{F}$ -Wiener process on $(\Omega, \mathcal{F}, \mathbb{P})$?
- (b) [10 marks] Let $(W(t))_{t \in [0, T]}$ be a Wiener process and let $T = 10$.

- i. Compute the following quantities:

$$\mathbb{E}[(W(4))^2], \quad \mathbb{E}[W(6)W(4)], \quad \mathbb{E}[W(6)(W(4))^2 - (W(4))^3].$$

- ii. Consider the function $g : [0, \infty) \rightarrow [0, \infty)$ given by $g(t) := 3 \cdot \mathbf{1}_{[2, 4)}(t)$. Find the distribution of the random variable

$$\int_0^{10} g(s) dW(s).$$

- (c) [8 marks] Let $(W(t))_{t \in [0, \infty)}$ be a Wiener process defined on the non-negative half-line. For $n \in \mathbb{N}$, consider the events

$$A_n = \{W(n+1) - W(n) > n\}$$

i. Show that for any $n \in \mathbb{N}$ we have

$$\mathbb{P}(A_n) \leq \frac{1}{n^2}.$$

ii. Show that with probability one, there exists only finitely many $n \in \mathbb{N}$ such that $W(n+1) - W(n) > n$.

(d) [9 marks] Let $(W(t))_{t \in [0, T]}$ be an $\mathbb{F}$ -Wiener process.

i. Find the stochastic differential of $(\cos(W(t)))_{t \in [0, T]}$

ii. Consider the process $(M(t))_{t \in [0, T]}$ defined by

$$M(t) := e^{\frac{t}{2}} \cos(W(t)).$$

By computing the stochastic differential of $(M(t))_{t \in [0, T]}$, show that it is an $\mathbb{F}$ -martingale.

(e) [8 marks] Let $(W(t))_{t \in [0, T]}$ be an $\mathbb{F}$ -Wiener. Let $\beta \in \mathbb{R}$ and consider the stochastic differential equation

$$dX(t) = \beta X(t) dt + dW(t), \quad X_0 = 0,$$

and let $(X(t))_{t \in [0, 1]}$ be its unique solution.

i. For $c \in \mathbb{R}$, find the stochastic differential of the process $(Y(t))_{t \in [0, T]}$, given by $Y(t) = e^{-ct} X(t)$, $t \in [0, T]$.

ii. Use your result above to find the solution $(X(t))_{t \in [0, T]}$ explicitly.

3. Let $T = 1$ and let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ satisfying the usual conditions. Let $(W(t))_{t \in [0, T]}$ be a one-dimensional $\mathbb{F}$ -Wiener process on $(\Omega, \mathcal{F}, \mathbb{P})$. Consider the Black-Scholes model where there exist exactly one riskless asset $(B(t))_{t \in [0, T]}$ and a risky asset $(S(t))_{t \in [0, T]}$. The dynamics of $(S(t))_{t \in [0, T]}$ is given by

$$dS(t) = 3S(t) dt + 2S(t) dW(t), \quad S(0) = 1, \quad t \in [0, T],$$

where $T > 0$ is the terminal time and the risk-free interest rate is given by $r = 1$.

(a) [6 marks] Let $\mathbb{P}^*$ be the risk-neutral probability measure of the above model and recall that under $\mathbb{P}^*$ the process

$$W^*(t) := \frac{3-1}{2}t + W(t)$$

is an $\mathbb{F}$ -Wiener process. Write down the stochastic equation (driven by W^*) that $(S(t))_{t \in [0, T]}$ satisfies under the measure $\mathbb{P}^*$ and write down the unique solution of this equation.

(b) [5 marks] Write down a differential equation for the riskless asset $(B(t))_{t \in [0, T]}$ and write down its unique solution provided that $B(0) = 1$.

(c) [10 marks] Consider a portfolio $H(t) = (H_1(t), H_2(t))$, $t \in [0, T]$.

i. When do we say that $(H(t))_{t \in [0, T]}$ is self-financing?

- ii. Suppose that $H_1(t) = 1$ and $H_2(t) = 1$, for all $t \in [0, 1]$. Find the expectation of the value of the portfolio $(H(t))_{t \in [0, T]}$ at the terminal time $T = 1$.
- (d) [9 marks] Consider a derivative written on the risky asset which pays at the terminal time $T = 1$, the amount $|S(1)|^8$ to the owner. Find the price of this derivative at time zero (give a real number and show your workings).

You can use that if $\theta \in \mathbb{R}$ and $X \sim N(0, 1)$, then the expectation of $\exp(\theta X)$ is given by $\exp(\frac{\theta^2}{2})$.